

BHUMIKA AGARWAL

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EXPERIENCE

JP Morgan Chase & Co.

Mumbai, India

Software Engineer II

Oct 2024 – Present

- Own the **overcharge dispute capability end-to-end** – from analysis through production – within an initiative modernizing the firm’s credit-card non-fraud **dispute-processing system**, building **Java task handlers on Camunda 7 (BPMN)** and **Spring Boot** services that resolve **10k+ dispute cases/month**.
- Analyzed legacy **Pega** dispute workflows – using **AI agents** to accelerate analysis – and helped re-platform them into a modern **Java + Camunda** architecture, translating existing business rules into reusable task handlers.
- Provisioned **AWS** infrastructure for the modernized dispute platform through **infrastructure-as-code** (an internal Terraform-based framework).
- Delivered **React + TypeScript** frontend enhancements on a bonds-trading platform, including an **event-tagging feature** that captures market events for historical tracking and market-trend analysis.
- Fixed backend defects (**Java 17**, Spring Boot, PostgreSQL) and raised automated **test coverage** across the bonds-trading platform.

Cognizant Technology Solutions

Mumbai, India

Software Engineer

Jul 2022 – Jul 2024

- Built **Spring Boot microservices and REST APIs** powering high-volume BFSI transaction workflows for enterprise banking clients.
- Tuned Hibernate/JPA query patterns and SQL execution plans across **20+** endpoints, **cutting average API latency by 30%** and reducing peak-hour database contention.
- Delivered reusable ReactJS dashboard components wired to backend services and strengthened JUnit and Mockito test coverage across service and component layers.

PROJECTS

Regulatory Approval System | *Java 17, Spring Boot 3, Camunda 7, Spring Security, JWT, Docker* | [Link](#)

- Engineered a BPMN-based regulatory approval platform on Camunda 7 using the **External Task Worker pattern** – 5 specialized workers handling risk scoring, compliance, escalation, notification, and completion – with **SLA enforcement (8h–48h)** and exponential-backoff retry.
- Synchronized JWT and **6-role RBAC** between Spring Security and Camunda Identity Service, enabling auditable workflow governance and enterprise-grade approval chains.

UrbanNexus | *Java 17, Spring Boot 3.5, React 19, TypeScript, MySQL, JWT, Docker* | [Link](#)

- Architected a multi-portal residential operations platform (Resident, Technician, SuperAdmin) with **role-isolated JWT authentication** across payment, booking, and operational workflows.
- Designed a transactional booking engine with automated technician matching, payment lifecycle orchestration, and JPA **@PrePersist** hooks for tax and audit-field automation.
- Shipped **25+** REST endpoints across a Spring Boot 3.5 + React 19 + TypeScript stack with audit trails, overdue-payment automation, and dual-mode H2/MySQL deployment.

TECHNICAL SKILLS

Backend: Java, Spring Boot, Camunda 7 (BPMN), Java Task Handlers, REST APIs, Microservices, Spring Security, Spring Data JPA, Hibernate, JWT, RBAC

Frontend: React, TypeScript, JavaScript, Material-UI, HTML5, CSS3

Databases: PostgreSQL, MySQL, Query Optimization

Cloud & DevOps: AWS, Infrastructure as Code (Terraform), Docker, GitHub Actions, Jenkins, Maven

AI & Testing: AI Agents(Claude, GitHub Copilot), JUnit 5, Mockito

EDUCATION

University of Petroleum and Energy Studies

Dehradun, Uttarakhand

B.Tech. in Computer Science Engineering

Jun 2018 – Jun 2022